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In situ variability of carrageenan content and biomass in the cultivated red macroalga *Kappaphycus alvarezii* with an estimation of its carrageenan stock at the scale of the Malasoro Bay (Indonesia) using satellite image processing

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Abstract The combination of in situ measurements, hydrodynamical modeling, and satellite imagery processing presents a complete tool to improve seaweed culture management. This study measured the evolution of carrageenan content during 1-year period and estimated the biomass of *Kappaphycus alvarezii* at a bay scale in Malasoro Bay, Indonesia. It allowed the determination of the carrageenan stock at the scale of the bay. The carrageenan content was assessed from different parts, i.e., basal and apical, of the thallus. The biomass from T_0 (beginning of the cultivation) to T_{45} (harvesting time) was determined at two seasons. Satellite image processing was performed to estimate the biomass at bay scale using parcels resulting from a semi-automatic delineation process. As no long-term in situ environmental data existed in the study area, a 3D hydrodynamical model (Model for Application at Regional Scale 3D) was implemented to obtain high-frequency salinity, water temperature, and currents. The high carrageenan yield was obtained from April to September 2015 (54–63%) when temperature ranged from 25 to 30 °C, salinity range from 33.8 to 34.8 psu, and precipitation below 0.5 mm. The biomass at bay scale was estimated at 2590 t with the highest carrageenan stock at 1.8 t ha^{-1} in May 2015. The carrageenan yields and quality obtained in this study fulfilled the specification recommended for industry, and the cultivation of *K. alvarezii* can be carried out at optimal environmental conditions on April–September. The comprehensive approaches used in this study provide information for carrageenan stock and seaweed culture management as an important economic activity to support Indonesian coastal communities.

Keywords Bay-scale monitoring · Biomass · Infrared analysis · Remote sensing · Carrageenan yield

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Introduction

Eco-friendly and organic products are a becoming trend as a business strategy for industries. Seaweeds, especially red algae, have attracted food industries, cosmetic, pharmaceutical, and biomaterial industries as an abundant edible source of high-quality sulfated polysaccharides and bioactive substances (Laurienzo 2010; Kim 2014; Wijesekara and Kim 2015). Among the family of commercially exploited polysaccharides, carrageenan has been widely used as the cheapest natural additive and ingredient hydrocolloid (Rhein-Knudsen et al. 2015). The global market demand of carrageenans, mainly kappa (κ), iota (ι), and lambda (λ), continues to grow rapidly because of their chemical and physical characteristics (Campo et al. 2009; Brenner et al. 2014), as a high-soluble dietary fibers that contribute to a wide variety of products in food and non-food industries (Necas and Bartosikova 2013;

Prajapati et al. 2014), especially as a gelling, emulsifying, stabilizing, suspending, and thickening agent.

In the last couple of decades, Southeast Asian region primarily Indonesia has led the production and export activity of the dry raw algal material and semi-refined carrageenan (SRC) (Hurtado et al. 2013). SRC is an insoluble residue containing important value of carrageenan and cellulose from the carrageenophytes. Carrageenan and other insoluble residues are obtained from the washing of seaweeds through a dissolution process with alkali solution and water (Periyasamy et al. 2016). The demand of carrageenan production is expected to increase constantly in the coming years. There are several critical conditions required for the red alga cultivation in order to accelerate its development in Indonesia, i.e., a vast suitable culture area, an affordable technology, a high availability of human resources, and government encouragement (Zamroni et al. 2011). In addition, the advantage of seaweed cultivation is its relatively small capital. Although production of carrageenophytes is increasing in recent years in Indonesia, its development has been hindered by a number of constraints, especially environmental conditions that influence carrageenan's biochemical variability and the appearance of diseases (MMAF 2015).

Environmental conditions such as water temperature, salinity, light, water motion, and nutrients are considered to influence polysaccharide biochemical contents (proteins, carbohydrates, lipids, fibers, ash, and nitrogen) (Cosson et al. 1995; Stengel et al. 2011; Tanniou et al. 2015; Stiger-Pouvreau et al. 2016). The effects are instantaneous, causing some serious problems in seaweed farming, i.e., oxidative stress which increases the occurrence of a serious disease called ice-ice, epiphyte outbreaks, and stunted individuals (Vairappan 2006). Face to these problems, the industrial good quality and the continuity of raw material supply in market have to be maintained and/or to be expanded. Red algae from the class Florideophyceae, especially the genera *Eucheuma* and *Kappaphycus*, are known for their richness in gelling carrageenans (Ask and Azanza 2002). The cultivation and the production of *Kappaphycus* such as *Kappaphycus alvarezii* have been recently optimized and their post-harvest processing are now standardized. Its carrageenan content has been extensively studied (Distantina et al. 2011; Iskandar et al., 2014; Wenno et al. 2015) but very few studies observed the temporal variability of this phycocolloid. Generally, in Indonesia, *K. alvarezii* is cultivated in a remote coastal area where lack of information on environmental data or biophysical parameter still remains.

In this context, tools like geographic information systems (GIS), remote sensing, and mapping can be helpful. The GIS has been used widely in all geographic and spatial aspects of seaweed management, including inventory and monitoring, zonation and farming site selection, planning, and environmental impact assessment (Rossi et al. 2010; Radiarta et al. 2011 ; Oppelt et al. 2012). Remote sensing allows monitoring of spatiotemporal distribution of macroalgae (Hu et al. 2015). Image satellite data from multispectral sensors with high or medium spatial resolution were widely used in mapping benthic macroalgae for biomass estimation (Andréfouët et al. 2004; Zubia et al. 2015), and along with airborne sensors, such as bathymetric Light Detection and Ranging (LiDAR), they are widely used to investigate and to characterize the macroalgal habitat and its environmental supporting parameters.

The objectives of this research were to study the in situ evolution of quality and quantity of carrageenan and to estimate the potential production of the cultivated *K. alvarezii.* Quality of carrageenan was represented by sulfate content and the quantity of carrageenan was determined by means of the carrageenan yield investigated at different thallus segments, i.e., apical and basal, in relation with the age. The potential production in terms of biomass and carrageenan at bay scale was estimated using a combination of in situ measurements of algal fresh weight production data and satellite imagery data. Environmental conditions (temperature, salinity, precipitation, and current) were also measured. Among all sites available for cultivation of red algae in Indonesia, we selected the Bay of Malasoro in South Sulawesi, which regroups all the good conditions for the cultivation of the selected species (Utojo et al. 2007).

Materials and methods

The study area

The study area was situated in a shallow semi-closed basin of Malasoro Bay (5° 36′ 46.40–5° 40′ 31.25″ S; 119° 35′ 14.88–119° 39′ 42.92″ E), South Sulawesi, Indonesia (Fig. 1)The province is known as a major commercial area for seaweed production in Indonesia (FAO 2014). Seaweed farms were generally situated on sandy or muddy-sandy substrates with depth ranges from 5 to 15 m. For the purpose of carrageenan study, thalli of *Kappaphycus alvarezii* were cultivated regularly from the same plot located at 119° 36′ 21.6″ E; 5° 38′ 47.76″ S (black cross on the map in Fig. 1a). In the study area, thalli of *K. alvarezii* were cultivated using traditional hanging floating long-line system. This system commonly uses sandbags as anchors and plastic bottles as floaters. The young, robust, and healthy branches (50 ± 5 g) were cut from the selected "mother plant" and were tied coupled in one knot for first seedling. At this bay, the studied algal species *K. alvarezii* is widely cultivated throughout the year and harvested every 40 ± 5 days (October 2014–November 2015).

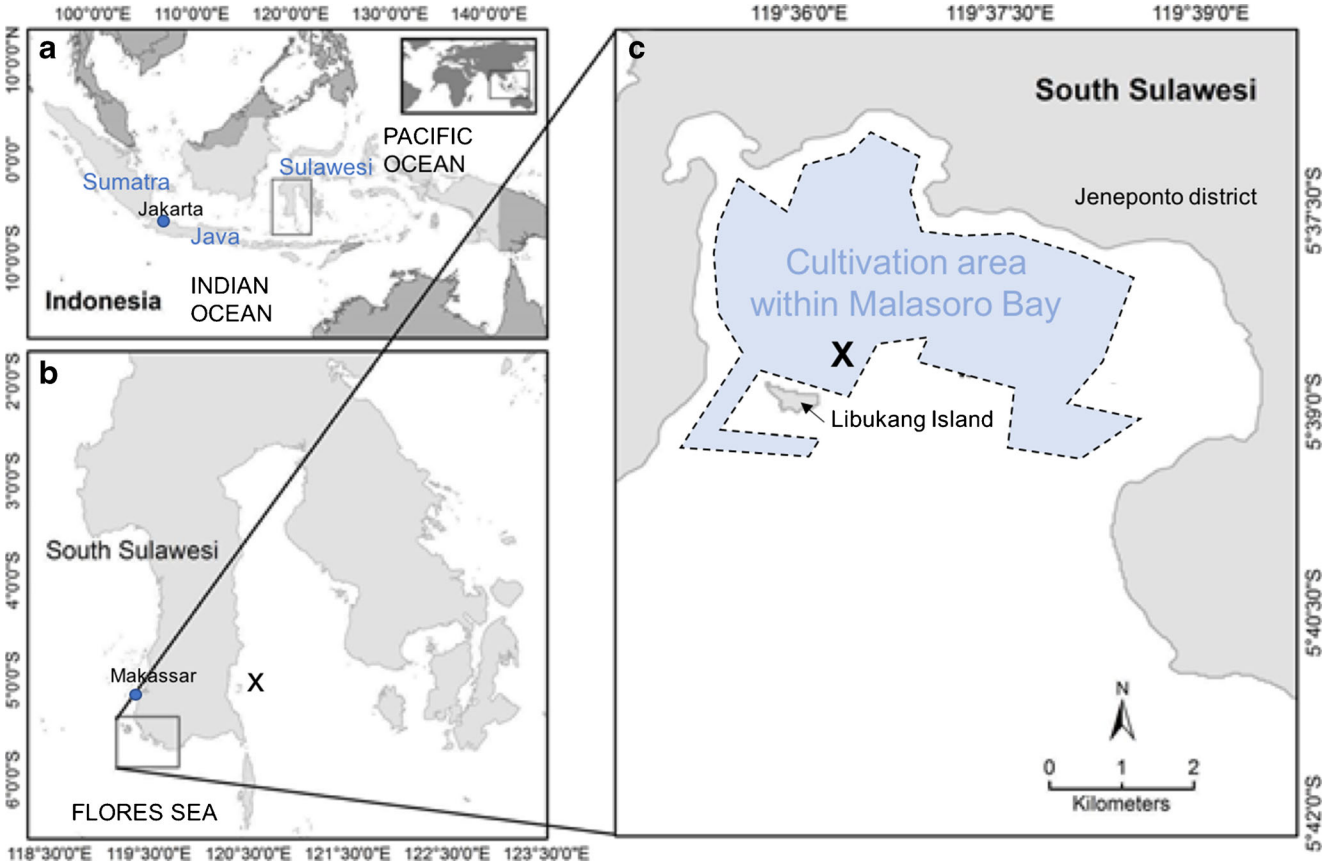

Fig. 1 Map of the study area. **a** General map of Indonesia in light grey, showing the area of study (cross). **b** Localization of the South Sulawesi Province. **c** General view of Malasoro Bay showing the area of cultivation of red algae (area surrounded by dotted line). *Black cross* on the map stands for sampling location (119° 36′ 21.6″ E; 5° 38′ 47.76″ S)

Environmental data

Collection of environmental data

As a tropical area, Indonesia is characterized by its monsoon climate and displays generally a well-marked seasonality. The same seasonality is observed at the scale of the study area in South Sulawesi. Precipitation is the main variable that influences the environmental conditions of the bay and occurs generally from November to March. Precipitation and wind velocity data were provided by Indonesian Agency for Meteorology, Climatology and Geophysics stationed near the study area in Jeneponto district—South Sulawesi (Indonesia).

This study was also associated with variability of climatic conditions in Indonesia, with alternance of oceanic/climatic phenomenons El Niño and La Niña. The Oceanic Niño Index (ONI) is a main measurement for monitoring, assessing, and predicting the strength of an El Niño/La Niña–Southern Oscillation (ENSO) episode. Period "El Niño" is characterized by a positive ONI greater than or equal to +0.5 °C while period "La Niña" is characterized by negative ONI less than or equal to −0.5 °C. The ONI is based on 3-month running mean sea surface temperature (SST) departures from average in the Niño 3.4 region (5° N–5° S and 170°–120° W). The SST reconstruction methodology is described in Huang et al. (2015). Map of bathymetry further in the manuscript will be used to enrich the hydrodynamic model. The environmental data are specifically presented in Table 1.

Hydrodynamical model

SST and salinity were recorded during all sampling periods to supply a hydrochemical model called "Model for Application at Regional Scale 3D (MARS3D)" licensed by Ifremer-France (Lazure and Dumas 2008). The hydrodynamic modeling tool used in this study is the MARS3D model. MARS is a very advanced code that syntheses knowledge of physical oceanography, mathematical modeling, and computer science. This 3D model has a mesh size of 50 m and is discretized in five vertical levels. Depth measurements used to build the bathymetric charts came from a navigation chart of Malasoro Bay edited by the Hydro-oceanographic Office of Indonesia. The lack of observational data led us to use data from large-scale models with spatial resolutions of several kilometers. Meteorological data were taken from ECMWF model from the Infrastructure Development of Space

Table 1 Environmental data during the period study, acquired in situ or determined by models

Parameters	Data source	Method
Sea surface temperature (SST)	3D hydrodynamical model	Model for Application at Regional Scale 3D (MARS3D) licensed by Ifremer-France and prepared by CEVA
Salinity	3D hydrodynamical model	Model for Application at Regional Scale 3D (MARS3D) licensed by Ifremer-France and prepared by CEVA
Precipitation	Daily precipitation (1 January 2014–31 December 2015) from Meteorological and Climate Bureau of Jeneponto, South Sulawesi	Monthly average of precipitation
Max current velocity	3D hydrodynamical model from CEVA	Model for Application at Regional Scale 3D (MARS3D) licensed by Ifremer-France and prepared by CEVA
Wind velocity	3 times measurement per day, provided by Meteorological and Climate Bureau of Jeneponto, South Sulawesi	Seasonality of wind velocity
Oceanic Niño Index/ONI using region Niño 3.4	NOAA Climate Prediction Center (January 2014–December 2015) (http://www.cpc.ncep.noaa.gov/data/indices/rindo_slpa.for)	

Oceanography (INDESO) portal site (http://www.indeso.web.id). This model has a spatial resolution of about 9 km. Physical data were extracted from ORCA model (a large-scale model) extracted and downloaded from the same portal site of INDESO project.

Study on the cultivated *Kappaphycus alvarezii*

Seaweed collection For the study on carrageenan, *K. alvarezii* was cultivated from 24 October 2014 to 18 November 2015. Each cultivation period was conducted for 45 days, except for the period from October to November 2014 and from April to May 2015 that took only 36 and 41 days, respectively, due to the appearance of disease (called ice-ice) infecting the cultivated seaweeds in the entire study area and also causing an epiphyte outbreak. The thalli of *K. alvarezii* were hand-collected, rinsed, and naturally sun-dried in respect with the steps used by Indonesian farmers. The algal biomass was kept in sealed plastic bags for further extraction and analysis on carrageenan yield and sulfate content (by Fourier transformed infrared (FTIR) analysis).

For the estimation of the algal biomass produced in the area, thalli of *K. alvarezii* were collected based on the age of the cultivar during periods in May and in November 2014. These periods correspond to the transition period from rain to dry season and from dry to rain season respectively. The weight per unit area in one cultivation plot was measured (see *Basic weight* under "Satellite image processing" section). Each sampling location was recorded using a GPS receiver. During the seaweed cultivation periods of May and November 2014, the growth rate of seaweeds was determined by weighing the seedling (W_i) and the adult individuals of *K. alvarezii* harvested after t days of cultivation (40 ± 5 days) noted as W_t, as described in Dawes et al. (1993).

Daily growth rate (DGR) of seaweeds was calculated and expressed as the percentage of increase in wet weight per day by the following equation described in Hung et al. (2009):

$$\mathrm{GR}\left(\%\ \mathrm{day}^{-1}\right) = \ln\left[\left(W_t / W_i\right) / t\right] \times 100$$

where W_i = average cutting wet weight at start, W_t = average cutting wet weight at time t, and t = time intervals (days). Weight gain of *K. alvarezii* in the area was also measured during the periods of May and November 2014 as the average of increase weight after 45 days of planting in comparison with the initial weight which was 50 g per individual.

Carrageenan extraction The method of carrageenan extraction used in this study was a modified method described in Stanley (1987). Seaweed material was washed thoroughly with tap water to remove salt, sand, epiphytes, and encrusting material, and then naturally sun-dried with the approximate 35% humidity. *Kappaphycus alvarezii* thalli regularly harvested within 1 year were separated into two different segments (apical and basal) to understand the effect of tissue age on carrageenan yield. The apical segments are characterized by young thalli parts (5–10 mm from the tips of the branches), while the old (mature) thalli of the axis are 30–50 mm from the basal segments (Mendoza et al. 2006). Furthermore, separation of apical and basal fragments could be applied into commercial requirement if desirable, as the apex is linked with a young growing thallus which is appropriate for seeding and the basal part is correlated with a potential harvestable part of the thallus.

For the extraction, 5 g of dried sampled *K. alvarezii* (DW) was cut into small pieces. Each sample was heated for 3 h in 6% KOH solution. This solution was pre-heated to 70 °C as an extractor which was immersed in a water bath and was set constant at 200 rpm. The constant ratio of seaweed weight to solvent volume (1:50; g:mL) was maintained by adding hot water. Filtrate was separated from residue using filter cloth and immediately poured into three volumes of cold (5 °C) technical ethanol (90% *w*), which caused precipitation of polysaccharides. The precipitation was done for 30 min with hand stirring. The precipitated carrageenans were collected and oven-dried at 50–60 °C to a constant weight. Experiments were carried out in triplicate. Dried carrageenan powder (W_c), obtained from extraction described above, was weighed and compared to seaweed raw material that was used at the beginning (W_i). Carrageenan content was expressed in percentage and calculated using the following equation (Góes and Reis 2012):

$$Carrageenan\ Yield(\%) = (W_c/W_i) \times 100.$$

FTIR analysis of carrageenan Extracted carrageenan chemical structure was assessed by FTIR spectroscopy (Chopin and Whalen 1993; Fournet et al. 1997). Carrageenan powder obtained from apical and basal segments of *K. alvarezii* were recorded using a Thermo Scientific Nicolet iS 5 iD 7 ATR accessory (a monolith diamond ATR type) in the 4000–400 cm^{-1} region. Spectra from FTIR were then analyzed for the carrageenan sulfate content. The result expressed as a degree of substitution (DS), which is the mean number of sulfate groups (moles) per main chain of disaccharide repeat unit in carrageenan (Rochas et al. 1986). The sulfate content of carrageenans was characterized by the following absorbance band ratios: 1250/2920 for total sulfate and total sugar content; 930/2920 for 3,6-anhydrogalactose-2-sulfate; and 845/2920 for galactose-4-sulfate and 805/2920 for 3,6-anhydrogalactose-2-sulfate (Pereira et al. 2009). The absorbance bands of apical and basal segments were detected at 2920 and 1210–1260 cm^{-1} regions (the S=O sulfate esters). Peaks were also observed at 912–940 cm^{-1} in all samples (3,6-anhydro-D-galactose). Another signal was recorded at 835–850 cm^{-1} bands, which is attributed to D-galactose-4-sulfate. Sulfates in the C4 position (the ring of galactose or D-galactose-4-sulfate) are associated with the rheological properties of carrageenan gels (Necas and Bartossikova 2013). All these signals were examined in all spectra obtained from freeze-dried samples of thalli analyzed by Fourier transform infrared spectroscopy with attenuated total reflectance. All the samples (apical and basal parts of thalli collected at different times of the cultivation period) were recorded in triplicates.

Satellite image processing

Biomass of *K. alvarezii* at bay scale was estimated using satellite image. This process went through three main steps. First, using the raw satellite image, parcels of cultivated *K. alvarezii* plot were created using a semi-automatic delineation process (Fig. 2). Secondly, an in situ field campaign was conducted to measure the parcels' basic weight at different times of seedlings (in concert with the algal farmer) on a specific known area in the bay. Basic weight was obtained by measuring the weight of thalli within a parcel. Thirdly, the total algal biomass of *K. alvarezii* was estimated at bay scale by multiplying the total area parcels by the mean biomass deduced from the field measurements, as shown in Fig. 2.

Seaweed parcel delineation

The satellite image exploited for this study was acquired by the GeoEye-1 on March 10, 2016 in standard mode 2A (image geometrically corrected). The image contained four spectral bands (1.8 m multispectral resolution that equals with distance sampling) at which values expressed in digital numbers (DN), see Table 2.

Perturbations of the light signal are observed on the raw satellite image in the four spectral bands due to atmospheric effects and turbidity because of light scattering from suspended particles. Sensor radiance data were converted to apparent reflectance using FLAASH, an atmospheric correction code based on the MODTRAN 4 and available in the ENVI image processing software. One of the main difficulties of underwater object detection, although the cultivated parcels are slightly submerged, is the influence of variable depth on the seaweed detection due to the radiance attenuation which is wavelength dependent. As no spectral library was collected at the time of satellite image acquisition and no information was available on water turbidity in this study, it was not possible to directly use the spectral signature of the seaweed parcels to enable their identification.

Finally, the strategy of parcel delineation was performed using successive multiresolution segmentations. The overall methodology can be summarized in the five following steps: (1) Land and clouds were masked. (2) A principal component analysis on the four spectral bands was applied for remaining seawater areas to obtain a single image highlighting the parcels to the background. (3) A Nagao image filter (Nagao and Matsuyama 1979) was applied to reduce noise while maintaining the parcel boundaries. (4) Successive multiresolution segmentations with decreasing scale factors were performed to identify rectangular shapes of parcels within the image. (5) The identified shapes that did not correspond to cultivated parcels were removed by visual interpretation.

Basic weight The long-line system of seaweed cultivation was composed by several parcels with ropes tying the seeds (Fig. 2). For each parcel, algae were sampled from five lines

(n) of 8.5-m line spacing (l), and 1.8-m distance of sampling (d) (Fig. 2). Parcel location was identified using a GPS receiver, and weight of algae sampled (w) was measured after 1 min of drying (Fig. 2). Time since seeding (t) was noted and should be the same for the n lines. Basic weight (BW) was then calculated using the following formula:

$$BW = \frac{n*w}{l*d}$$

with n = number of line sampled; w = weight of algae sampled; $l = n$ line spacing; and d = distance of sampling.

Biomass estimation The relationship between the two measured variables time since seeding (TSS) and BW was fitted using an exponential model. The total biomass at the entire bay was assessed by multiplying the mean basic weight deduced from in situ measurements with the total surface of the delineate parcels. The mean basic weight was estimated by

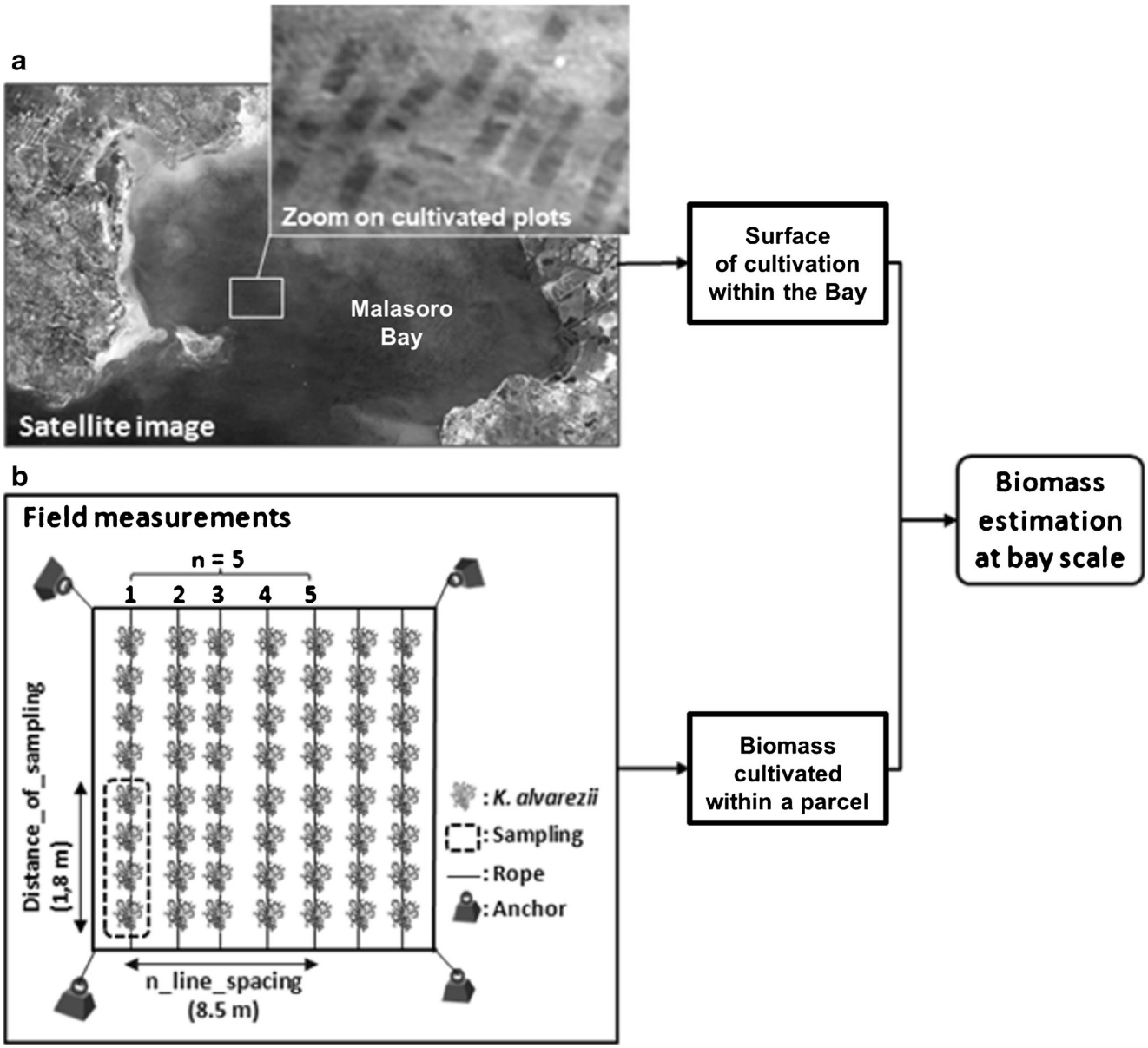

Fig. 2 Schematic procedure showing the two steps followed for the estimation of the cultivated biomass of *Kappaphycus alvarezii*, at the station black cross (see Fig. 1) within Malasoro Bay (South Sulawesi, Indonesia). **a** Satellite images of cultivated plots in the study area. **b** Schematic aerial view of a floating long-line cultivation parcel of *K. alvarezii* showing the hanging sampled thalli (dotted lines). The estimation of the basic weight was obtained by selecting randomly five lines in a parcel and by measuring the weight of four thalli by line, collected randomly within parcels

Table 2 Spectral band description of the satellite image (FWHM = full width at half maximum)

Wavelengths (nm)	FWHM (nm)	Description
480.0	58.4	Blue
545.0	64.6	Green
672.5	31.6	Red
850.0	101.2	Near infrared

using the exponential model at the day corresponding to the average distribution of the time since seeding.

Estimation of the carrageenan production at the scale of the bay

The estimation of the carrageenan productions at the scale of the entire bay and considering both seasons were simulated, using the following methodology: (i) the estimation of total biomass was done as basic or fresh weight of *K. alvarezii* at bay scale, (ii) the total dry biomass at bay scale (DW) of *K. alvarezii* was obtained from multiplying the fresh weight at bay scale by standard requirement of raw dried seaweed from carrageenan companies and the National Standardization Agency of Indonesia using an upper limit to moisture content of 35% (Mulyati and Geldermann 2017), (iii) the carrageenan production of *K. alvarezii* was simulated exclusively using the result of the total biomass at bay scale and carrageenan yield obtained (mean of apical and basal fragments) during the study period (see Table 3). Knowing the cultivated surface area in the bay, we were then able to give a quantity of carrageenan in relation with the cultivated area, expressed in t ha^{-1}.

Statistical analysis

The carrageenan contents and the degree of substitution of sulfate content were expressed as mean ± SD (standard deviation) from triplicate measurements for each sample. Results were analyzed statistically (one-way ANOVA, Tukey's test) at 95% of significance using IBM SPSS ver. 20 software to detect significant differences among the target species collected at different stages and also in different seasons. Pearson's correlation analysis was used to determine the correlation coefficients among biological data and environmental parameters (SST, sea surface salinity, precipitation, and carrageenan yield based on branch part of *K. alvarezii*).

Results

Carrageenan content, daily growth rate, and weight gain

Yield and quality Carrageenan yield on the cultivated *K. alvarezii* samples varied with periods of cultivation (Fig. 3a). One should note a carrageenan yield more important during high season, from April to November 2015 (Fig. 3a). Then, two distinct seasons were observed: (1) high-yield period: season with high carrageenan yields (end of April to September) reaching almost 60%, and (2) low-yield period: season with low carrageenan yield around 46% (October to early April). This species thus produced more carrageenan during the dry season than during the rainy season.

The carrageenan yields also varied with the part of the thallus, considering young (apical) and mature (basal) segment branches as demonstrated in Table 3. During high season, higher carrageenan yields were obtained from young thalli and vice versa, and mature thalli produced more carrageenan during low season.

Table 3 Carrageenan yield (CY) from different types of fragments, apical and basal, of *Kappaphycus alvarezii*, production of carrageenan at bay scale (tons), production of carrageenan (t ha^{-1}), and mean of environmental parameters (SST, salinity, and precipitation) during the study period. The minimal and maximal yields are visible by the mention of min and max

Period	CY from apical (young) ± SD (%)	CY from basal (mature) ± SD (%)	Whole thallus ± SD (%)	Carrageenan at bay scale (t)	Carrageenan related to area (t ha^{-1})	Mean SST (°C)	Mean salinity (psu)	Precipitation (mm)
Oct.–Nov. 2014	50.9 ± 3.3	52.6 ± 0.9	51.8 ± 2.1	469.1 ± 18.7	1.5	28.5	34.7	83.7
Nov.–Dec. 2014	48.2 ± 0.4	49.0 ± 1.7	48.6 ± 1.0	440.4 ± 9.5	1.4	30.0	33.6	1004
Jan. –Feb. 2015	41.9 ± 2.4	50.8 ± 0.9	46.3 ± 4.5	420 ± 40.5	1.4	29.7	33.0	486.4
Mar.–Apr. 2015	42.5 ± 1.4	45.8 ± 1.9	44.2 ± 2.0 (min)	400.4 ± 18.3	1.3	29.7	33.1	257.2
Apr.–May 2015	62.4 ± 2.2	60.4 ± 3.4	61.0 ± 2.5 (max)	553.1 ± 22.5	1.8	28.8	33.8	0
May–Jun. 2015	61.7 ± 1.4	57.8 ± 1.2	59.8 ± 2.0	541.8 ± 18.3	1.7	27.2	34.2	0.5
Jul.–Aug. 2015	59.9 ± 2.4	54.8 ± 1.3	57.4 ± 2.8	520 ± 25.6	1.7	25.8	34.6	0
Aug.–Sept. 2015	48.9 ± 1.7	63.3 ± 2	56.1 ± 7.2	508.5 ± 65.0	1.6	25.6	34.8	0
Oct.–Nov. 2015	42.6 ± 1.3	54.8 ± 3.3	48.7 ± 6.1	441.5 ± 55.3	1.4	27.4	34.8	0

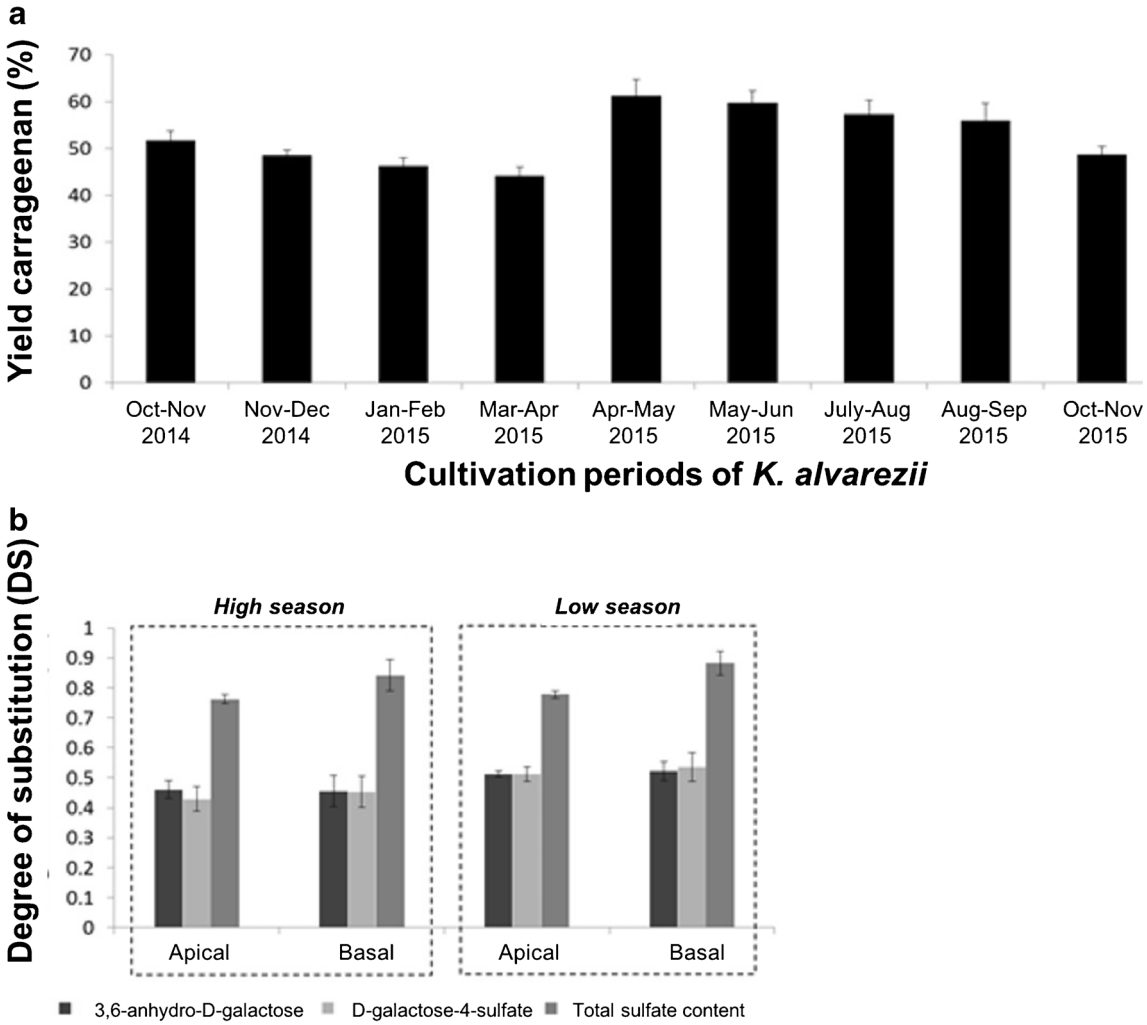

Fig. 3 Carrageenan produced by *Kappaphycus alvarezii* cultivated in Malasoro Bay (Indonesia). **a** Seasonal variation of carrageenan yields all along the study period. **b** DS index of carrageenans produced by *K. alvarezii* in relation with the different parts, i.e. basal or apical, of the thallus obtained from spectral data using Fourier Transformed Infra-Red analysis

In young thalli, the highest value of carrageenan yield (62.4 ± 2.2%) was obtained in April–May 2015, while the lowest value (41.9 ± 2.4%) was obtained in January–February 2015. In mature thalli, the highest value of the yield (63.3 ± 2%) was obtained in August–September 2015, while the lowest value (45.8 ± 1.9%) was observed in March–April 2015.

Based on the FTIR analysis on carrageenan powder, *K. alvarezii* is considered as a main source of kappa carrageenan, as seen by the signal on the FTIR spectrum: 1242, 1072, 933, and 847 cm^{-1} (data not shown). The total sulfate content in the young and mature thalli, represented by DS at apical and basal carrageenan yield, was viewed by the signal ratio 1250/2920 cm^{-1}, and this sulfate content is higher in basal thalli than apical ones (Fig. 3b). Nevertheless, this difference is not significant (one-way ANOVA, $p = 0.05$).

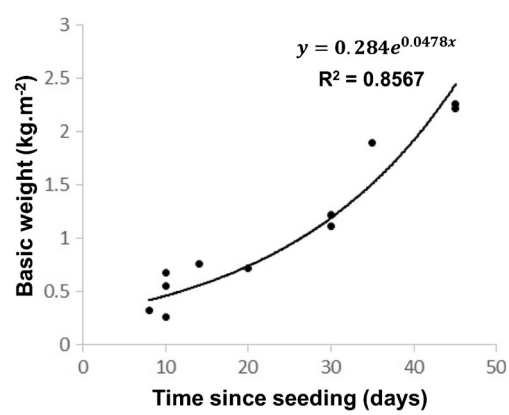

Fig. 4 Relation between the basic weight (BW) and the time since seeding (TSS) regrouping data obtained during the field campaign carried out in May 2014 on the cultivated red macroalga Kappaphycus alvarezii in the Malasoro Bay (Indonesia)

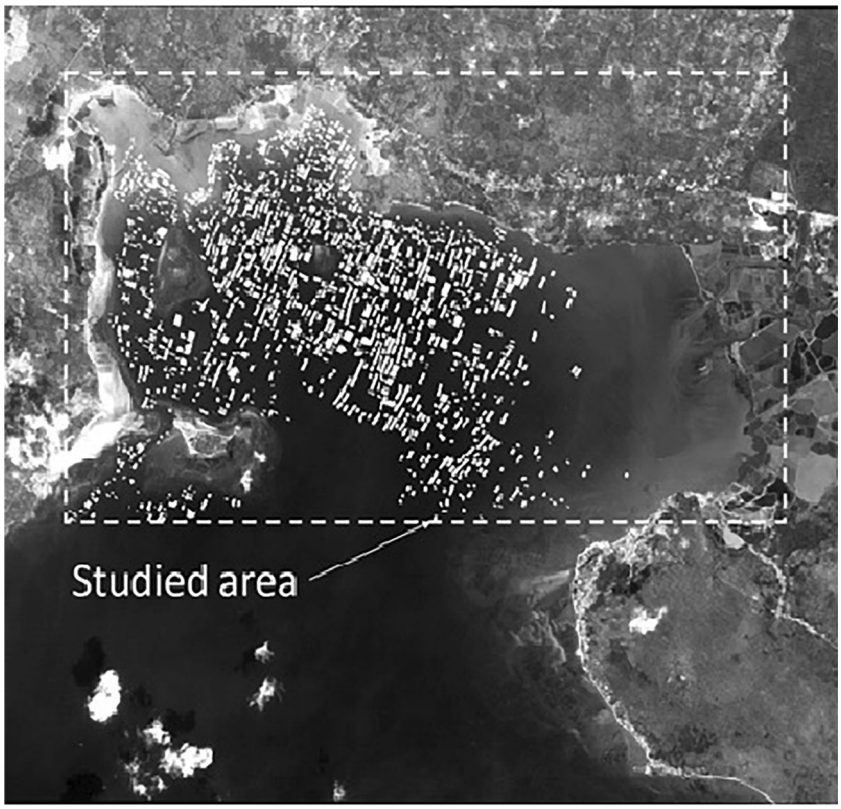

Fig. 5 Locations of parcels of the cultivated *Kappaphycus alvarezii*, identified by the semi-automatic delineation process within the study area (Malasoro Bay, Indonesia)

Daily growth rate and weight gain The measurement of growth in the study area conducted in the middle of the study period, May and November 2014, permitted to highlight a high growth rate of 4.3% day^{-1} in May (high production period with low temperature), and a low growth rate of 2.8% day^{-1} in November (low production period with high temperature). The weight gain of cultivated seaweeds, after 45 days of planting, was also higher in May (238.2 ± 38.4 g) than in November (179.4 ± 32.7 g).

Assessment of the biomass production at the scale of the bay

Basic weights from in situ measurements BW and TSS were measured in the study area on May 2014 (Fig. 4). The relationship between these two parameters was fitted using an exponential model with a coefficient determination of $R^2 \approx 0.86$ (see Fig. 4). The resulting exponential model was

$$BW = 0.284e^{0.0478*TSS}$$

with BW = basic weight (fresh weight), in kg m^{-2} and TSS = time since seeding, in days.

Biomass estimation from seaweed parcel delineation The semi-automatic delineation process led to the identification of 1642 parcels for a total culture surface of about 311.1 ha (see Fig. 5). Parcels, as seen in white on Fig. 5, are present in the western part of Malasoro Bay. The average distribution of TSS value was 22.5 days corresponding to a basic weight of 0.8325 kg m^{-2} by using the exponential model approximation. The biomass in fresh weight of *K. alvarezii* for the whole bay was estimated at 2590 t. This result has to be considered as a first assessment of the total biomass because this approach did not take into account the biomass heterogeneity of each seaweed cultivated parcel (i.e., the parcel basic weight at different times of seedlings).

Estimation of the carrageenan production for the bay

Considering the 35% moisture content, the dry raw material that could be obtained at the scale of the bay was estimated at 906.5 t. From the carrageenan production obtained from the dry weight of whole thalli and the yield determined from the mean of apical and basal fragments (Table 3), the highest value of carrageenan production was 553.1 t obtained in May 2015, while a lowest value was determined at 400.4 t in February 2015. The production of carrageenan for each harvest period is presented in Table 3.

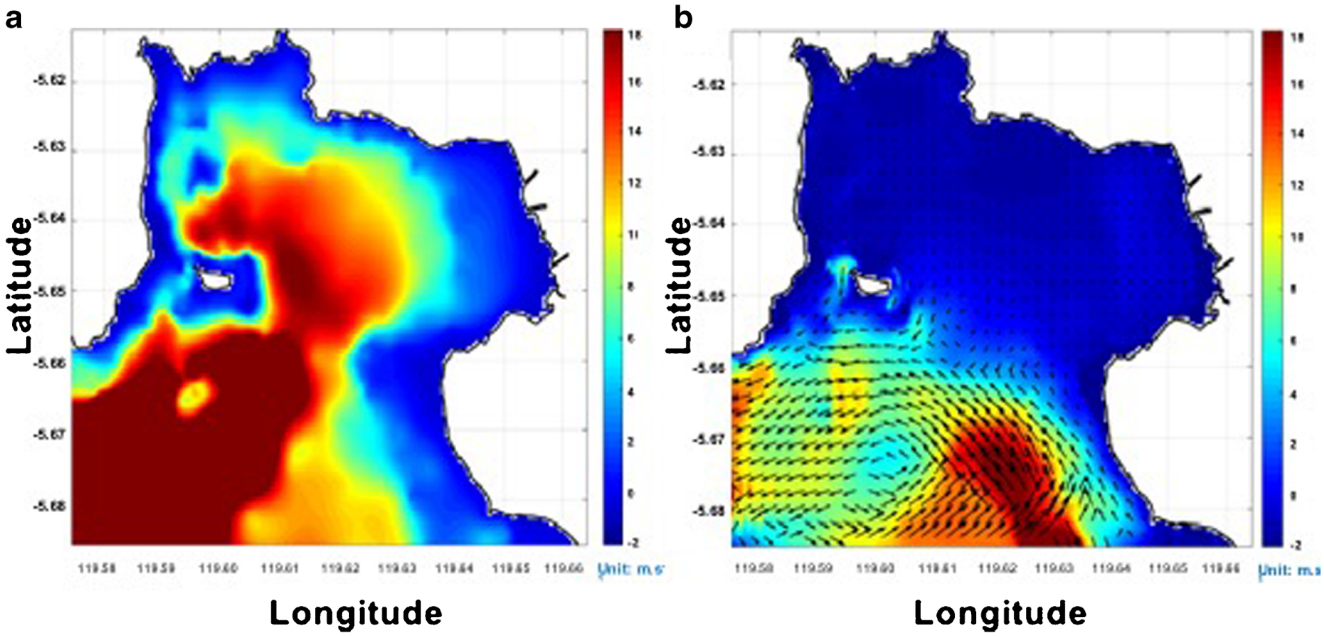

Fig. 6 Physical parameters in the Malasoro Bay determined by the model MARS3D. **a** Bathymetric chart of Malasoro Bay in corrected meters below mean sea level—*red color* shows the deepest part of the bay and *blue color* the shallowest part of the bay. **b** Maximum velocity field obtained from the model (see Table 1)—*red color* shows the location of the strongest currents (m s^{-1}) while the weakest are located in the *blue* part. *Arrows* show the direction of the currents

Environmental conditions within the bay

Maximum instantaneous velocity computed by the 3D hydrodynamical model on the whole month of September 2014 is presented in Fig. 6. It showed an apparently homogenous vector field, which generally oriented to the ebb direction. The maximum currents, about 0.5 m s^{-1} (Fig. 6b), were located at the entrance of the Malasoro Bay and were considered as

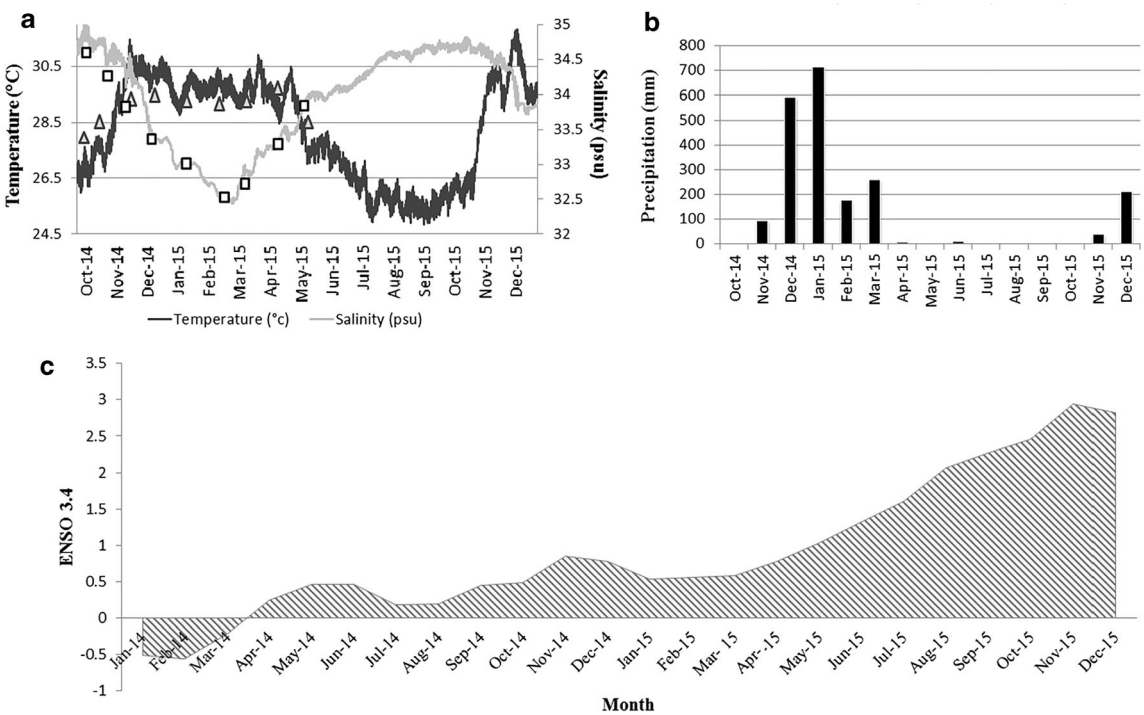

Fig. 7 Environmental conditions at the Malasoro Bay during the study period. **a** SST and salinity were obtained from simulation using MARS3D hydrodynamical model at a fixed location (119.606 °E/ −5.646° N). The *triangular* and *square symbols* represent in situ data of temperature and salinity, respectively. **b** Monthly average precipitation covering the study period. **c** ENSO 3.4

Table 4 Pearson's coefficients of correlation (*r*) between temperature (°C), salinity, precipitation (mm), and carrageenan yield (CY) for *Kappaphycus alvarezii* cultivated during 1 year in Malasoro Bay (Indonesia). Significant *r* values are in italics

	SST (°C)	Salinity (psu)	Precipitation (mm)	CY-apical (%)	CY-basal (%)
SST (°C)	1				
Salinity (psu)	−0.772	1			
Precipitation (mm)	0.645	*−0.865*	1		
CY-apical (%)	−0.756	0.647	*−0.698*	1	
CY-basal (%)	*−0.804*	0.668	−0.509	*0.858*	1

weak. At the bottom where the cultivations were settled, the current was globally inferior to 5 cm s^{-1}. The conditions of temperature, salinity, and precipitation in the study area are presented in Fig. 7a. Temperature and salinity predicted by the 3D model were consistent with in situ data (Fig. 7a). The monthly mean temperature ranged from 25 °C in September to 30 °C in December with fluctuations throughout the year of about 5 °C (Table 3 and Fig. 7a). Mean salinity ranged from 33.0 psu in December to 34.8 psu in September (Table 3). In general, salinity follows inversely the pattern of temperature. The fluctuation of salinity in the study area seemed to be influenced by precipitation (Fig. 7a, b). The peak of precipitation was observed in December (1004 mm). Annual precipitation below 1000 mm $year^{-1}$ was categorized as low precipitation. Dry months (precipitation around 0 mm) were observed from April to September (Fig. 7 and Table 3). ENSO index showed a stronger variation at the end of the study period, from September to December 2015, with an index above 2.5 (Fig. 7c).

Analysis of correlation coefficient among precipitation, SST, salinity, and yield of carrageenan shows that SST influenced the carrageenan yield stronger than salinity (Table 4). In general, these correlations were slightly stronger on basal segment than apical segments (0.8 for SST and 0.6 for salinity). SST correlates negatively with carrageenan yields, while salinity correlates positively.

Discussion

This study analyzed temporal variations of carrageenan yields and sulfate contents in *K. alvarezii* in Malasoro Bay, Indonesia. In order to increase the quality and quantity of carrageenan, it is essential to understand the environmental factors for growth of *K. alvarezii*. Environmental data such as temperature, salinity, and precipitation were incorporated to describe these variations related to season. The stock of *K. alvarezii* at the scale of a bay was also estimated using satellite imagery and in situ measurements, which gives crucial information for the management of carrageenophyte cultivation in Indonesia.

Carrageenan yields and seasonal variability Carrageenan yields of *K. alvarezii* from apical and basal segments were different. In high season (with high carrageenan production: April–September), high carrageenan yields were obtained in apical segments than in basal segments, while in low season (with low carrageenan production: October–March), higher carrageenan yields were obtained in basal segments than in apical ones (Table 3). Delattre (2011) mentioned that the nature of carrageenan produced by macroalgae such as *K. alvarezii* depends on age. Moreover, Mendoza et al. (2006) stated that carrageenan yield was relatively higher in the apical (young) segments than in the basal (mature) segments. Other studies in a stable water condition showed similar results and further mentioned that low carrageenan yield in basal segments was explained by the maturity of thalli (John and Asare 1975; Reis et al. 2008). Carrageenans are located in the algal cell wall as an amorphous and hydrosoluble matrix. The κ-carrageenan in the thallus of *K. alvarezii* is found mainly in medullary cell walls (Zablackis et al. 1991). Medullary cells seemed larger in the axes as observed with the mean diameter profiles reflecting a maturation of the tissues (Lechat 1997). Mendoza et al. (2006) stated that physiological and structural functions during the growth of *Kappaphycus striatum* may control the biosynthesis of sulfated galactans.

The hydrodynamical model for regional scale, MARS3D used in this study, could figure out the abiotic factors, i.e., temperature, precipitation, and hydrodynamical conditions, in semi-closed bay. Temperature values obtained from the model showed the same range, 27.3–29 °C, in a previous study carried out in Malasoro Bay (Utojo et al. 2007). Our results showed also the correlation between the abiotic factors from the model and the carrageenan yields from apical and basal segments, growth rates, and weight gain. Meanwhile, carrageenan quality, such as gelling characteristic using sulfate content in this study, is mostly related to the duration of cultivation, harvesting time, post-harvest handling, and extraction method as demonstrated by some previous works (Muñoz et al. 2004; Hayashi et al. 2007; Góes and Reis 2012; Periyasamy et al. 2016).

Temperature seems to be one of the main environmental factors affecting the carrageenan yield. In our study, high carrageenan yields (56–62%) were obtained from specimens collected in low water temperature (25.5–28.5 °C) in April–September, while low carrageenan yields (45–52%) were obtained in high water temperature (28.5–30.0 °C) in October–March (Table 3). During low seawater temperature, thalli of *K. alvarezii* were also healthier and possessed darker coloration than during the

Table 5 Comparison of the present study with literature in terms of carrageenan yield, temperature, and period of cultivation of *Kappaphycus alvarezii* from different locations

No.	Sites	High carrageenan yields (temperature; months)	Low carrageenan yields (temperature; months)	Source
1	Malasoro Bay, Indonesia	56–62% (low temperature: 25–28 °C; April–September)	45–54% (high temperature: 28–30 °C; October–March)	This study
2	Cam Rahn Bay, Vietnam	45–49.8% (low temperature: 24–26; September–February)	40.9–42.2% (high temperature 28–31.5; March–August)	Hung et al. 2009
3	Panguil Bay, Philippines	67.2–79.62% (low temperature: 27–30 °C; June–September)	22–30.5% (high temperature: 32 °C; April–May)	Orbita 2013
4	Marambia Bay, Brazil	61.7% (high temperature: 28 °C; January)	38–45% (low temperature: 22 °C; August–September)	Goes and Reis 2012
5	Palk Bay, India	37–38% (high temperature: 32–35 °C; March–September)	35.3–37% (low temperature: 28.3 °C; October–February)	Periyasamy et al. 2016

other period (Setyawidati, pers. observ.). Our present study indicates then that the lower sea surface temperature (SST) all along the year was suitable for the implementation of commercial cultivation in the study area.

Table 5 shows the range value of yield carrageenan in this study in comparison with other studies. Orbita (2013) working in the Philippines also found a similar result that high carrageenan yields occurred in low water temperature during the period June–September and low carrageenan yields occurred in high temperature during the period from October to May. Another study conducted in Vietnam by Hung et al. (2009) also found similar results (Table 5). In contrast, different patterns of results, always on *K. alvarezii*, were reported in India (Periyasamy et al. 2016) and Brazil (Góes and Reis 2012), where high carrageenan yields occurred during periods with high temperature.

Based on Table 5, it was found that carrageenophytes may need favorable and optimal temperature for allowing their best growth rates and carrageenan yield range in different locations. Temperature is one among other hydrodynamical conditions affecting the variability in carrageenan yield in different locations (Glenn and Doty 1990). However, several authors (Percival 1979; Durako and Dawes 1980; Kloareg and Quatrano 1988) hypothesized that carrageenophytes produce more carrageenan to support local stress. This was also mentioned by Orbita et al. (2013) for *K. alvarezii* studied in the Philippines. In another species, *Hypnea musciformis*, Reis et al. (2008) arrived at the same conclusion concerning the positive effect of hydrodynamical conditions on the carrageenan yield. The different impacts of temperature on growth rates and carrageenan yields among studies suggested the influence of season and, more interestingly, the influence of geographic locations as mentioned also by Periyasamy et al. (2016). Indeed, similar pattern was found among studies conducted in the Southeast Asia region, i.e., Indonesia, Vietnam, and the Philippines, while different patterns were found in studies conducted in other geographical areas, like Atlantic (Brazil) and Indian (India) Oceans.

The measurement of growth in the study area conducted in the middle of the study period, May and November 2014, permitted to highlight a high growth rate of 4.3% day^{-1} in May (high production period with low temperature) and low growth rate of 2.8% day^{-1} in November (low production period with high temperature). The high growth rate in this study was above the recommended commercial value of 3.5% day^{-1} for eucheumoid farming (Ask and Azanza 2002). The weight gain of cultivated seaweeds (after 45 days of planting) was likewise higher in May (238.24 ± 38.44 g) than in November (179.39 ± 32.67 g). High temperature seems to inhibit the growth of fragments and to limit the chance for survival as mentioned also by Araujo et al. (2014).

In addition, low temperature from April to September coincides with the Southeast Asia-Australia monsoon southeasterly wind from Australia. It generates upwelling and brings cooler and nutrient-rich water near the surface, enhancing productivity and increasing ocean color in Indonesian seas (Susanto et al. 2006). Local climatic conditions, precipitation in Indonesia are also closely related to ENSO. Low levels of precipitation and upwelling that brings cooler waters and increases nutrients are the phenomena generated by El Niño. In the opposite, La Niña causes high level of precipitation. Furthermore, El Niño and La Niña controlled the tropical ocean productivity (Susanto and Marra 2005), which later affect the productivity of seaweed cultivation. More precisely, Radiarta et al. (2013) proved that the local and global climate change determined the seaweed cultivation periods in Gerupuk Bay, Lombok, Indonesia. The extreme monsoon events, such as El Niño or La Niña, could also influence carrageenan production (Ask and Azanza 2002; Araujo et al. 2014). ONI is the method selected in our study to evaluate climate anomaly (El Niño and La Niña). Based on the result obtained, the occurred phenomena during this study was mostly El Niño particularly from May 2015 to the end of

our study, which was reinforced by the low level of precipitation. In our study, high carrageenan yields were obtained in warmer months, like from April to September. High carrageenan yields in October–November 2015 were probably related to ENSO 3.4, which prolonged the duration of warmer conditions (Figs. 3 and 7). Other abiotic factors could influence carrageenan yields of *K. alvarezii* along our study period, as salinity and precipitation. Salinity is known to influence carrageenan yield and quality since carrageenan is responsible for the ionic equilibrium of the cell (Percival 1979). Salinity ranges during high carrageenan production were 33.8–34.8 psu, similar to the one observed in low season (33.0–34.0 psu), and precipitation during high production period was lower (0.0–0.5 mm) than the one during the low production period (0.0–1004.0 mm). The amplitude of salinity noted during our study period was not enough to explain the variability in the carrageenan yield, which was not the case of precipitation which highly varied between both seasons.

From the FTIR analysis of carrageenans extracted in this study, we were able to characterize the quality of the phycocolloid from *K. alvarezii*, as type of κ-carrageenans and their sulfate content. Index of total sulfate content showed no difference of absorbance ratio between apical and basal segments during the study period (Fig. 3b). As for 3,6-anhydro-D-galactose and D-galactose-4-sulfate, DS index between apical and basal segments varied throughout the study period.

Carrageenan stock The satellite image processing with high spatial resolution allowed an indirect assessment of the algal biomass at the bay scale of Malasoro, Indonesia. In our present study, the stock from extrapolation for biomass in the entire bay was estimated at 2590 t (fresh weight). The current methodology for assessing total biomass of *K. alvarezii* by remote sensing can be improved by taking into account the basic weight of each parcel at different times of seeding. Computing a vegetal index on satellite image would enable this parameter to be taken into account in order to better estimate the total seaweed biomass. This approach involves taking into account the signal attenuation due to the water column and suspended sediment and to collect field spectral data.

Based on three approaches to simulate the carrageenan production per hectare, the highest (66.3%) and lowest (38.6%) carrageenan yields were used. Therefore, the highest and the lowest biomasses of carrageenan per unit area were estimated at 1.8 and 1.3 t ha^{-1}, respectively (Table 3). In a previous study, a yearly total output carried out on eucheumoids at Bali (Indonesia) was estimated at approximately 2500 t dry weight and occupied 75 ha of reef-flat area (Adnan and Porse 1987).

Despite hypotheses taken to assess algal biomass, our study is the first to give an estimation of the carrageenan production at a bay scale, and the results provide basis information for other similar studies or for studies using different satellite image sources. A satellite image efficiently allows the assessment of resources, but further studies are needed to better estimate and better quantify the natural temporal variability among habitats. The presence of other object(s) in the study area may blur the signal of the studied algae. Similar strategy of methodology with multi-data of satellite sensor is then suggested.

In our present study, it was assumed that only one species was cultivated in this bay, following discussions with farmers within the Bay of Malasoro (Setyawidati, pers. com.). Moreover, the type of sea bottom was quite homogenous, and water turbidity and the small number of in situ data drove us to remove the part concerning the imagery processing to assess the algal biomass. The satellite image data could be used not only to delineate the parcels but also to accurately estimate and monitor the total biomass by computing indexes such as the SVCI index in order to take into account the biomass heterogeneity of each parcel (parameter time since seeding). One possibility will be to take into account the signal attenuation due to the water column and suspended sediment and to collect more field spectral data.

Some studies employed a multispectral image satellite calibrated with in situ measurements to assess biomass in natural populations and cultivated seaweed areas. For example, Andrefouet et al. (2004) used field data and three 4 m resolution IKONOS satellite image to map and to estimate the total biomass of Sargassaceae species, i.e., *Turbinaria* and *Sargassum* genera, invading coral reefs from French Polynesia (Stiger and Payri 2005). Huo (2015) investigated the compositions, changes, and biomasses of attached *Ulva* species on *Porphyra* rafts along the radial sandbank in the Yellow Sea and their potential contributions to green tides using a combination of oceanographical condition data, biomass transect, and image satellite analysis. In our study, we combined the ability of remote sensing (to estimate the parcels' basic weight), in situ measurements (to estimate the biomass in relation to parcels at different time seedlings), and laboratory work (for the carrageenan extraction and yield), to assess the carrageenan production at bay scale. This will provide information to help managers and allow them to better and sustainably exploit algal resources in the area.

The carrageenan yields obtained in this study (41.9–63.3%) fulfilled the specification recommended for industry which was ±38% from Orbita (2013). Thus, the cultivation of *K. alvarezii* in the study area is promising. In addition, the macroalgal biomass is an important contributor to the primary productivity in the study area and *K. alvarezii* can be cultivated throughout the year (temperature ranged from 25 to 30 °C), especially in April–September when temperature became optimal for carrageenan production. The almost stable salinity conditions in the study area that ranged from 33.8 to 34.8 throughout the year also support the seaweed cultivation.

Planting and harvesting during optimal conditions could optimize aquaculture production and could avoid crop failure, as well as developing a sustainable seaweed aquaculture in Indonesia.

This study provides the basic information and early assessment that could help to manage the carrageenan stock of *K. alvarezii*, a source of commercial kappa carrageenan in the global market. Our results encourage carrageenan production, not only in Malasoro Bay but also in Indonesia, as our data should be taken into consideration for the management of *K. alvarezii* cultivation. In situ measurement data, hydrodynamical model combined with satellite image can be applied to acknowledge the production volume in seaweed aquaculture and to monitor the development of seaweed cultivation for Indonesia. The high quality and production from local cultivations of *K. alvarezii* and the indigenous manufacture of carrageenan have a way for potential export to other countries and thus open a pathway for the livelihood of considerably poor coastal Indonesian fisher folk.

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References

Adnan H, Porse H (1987) Culture of *Eucheuma cottonii* and *Eucheuma spinosum* in Indonesia. Hydrobiologia 151/152:355–358

Andréfouët S, Zubia M, Payri C (2004) Mapping and biomass estimation of the invasive brown algae *Turbinaria ornata* (Turner) J. Agardh and *Sargassum mangarevense* (Grunow) Setchell on heterogeneous Tahitian coral reefs using 4-meter resolution IKONOS satellite data. Coral Reefs 23:26–38

Araújo PG, Ribeiro ALNL, Yokoya NS, Fujii MT (2014) Temperature and salinity responses of drifting specimens of *Kappaphycus alvarezii* (Gigartinales, Rhodophyta) farmed on the Brazilian tropical coast. J Appl Phycol 26:1979–1988

Ask EI, Azanza RV (2002) Advances in cultivation technology of commercial eucheumatoid species: a review with suggestions for future research. Aquaculture 206:257–277

Brenner T, Tuvikene R, Parker A, Matsukawa S, Nishinari K (2014) Rheology and structure of mixed kappa-carrageenan/iota-carrageenan gels. Food Hydrocoll 39:272–279

Campo VL, Kawano DF, da Silva Jr DB, Carvalho I (2009) Carrageenans: biological properties, chemical modifications and structural analysis—a review. Carbohydr Polym 77:167–180

Chopin T, Whalen E (1993) A new and rapid method for carrageenan identification by FTIR diffuse-reflectance spectroscopy directly on dried, ground algal material. Carbohydr Res 246:51–59

Cosson J, Deslandes E, Zinoun M, Mouradi-Giverneau A (1995) Carrageenans and agars. Prog Phycol Res 11:269–324

Dawes CJ, Trono GC Jr, Lluisma AO (1993) Clonal propagation of *Eucheuma denticulatum* and *Kappaphycus alvarezii* for Philippine seaweed farms. Hydrobiologia 260:379–383

Delattre C, Fenoradosoa AT, Michaud P (2011) Galactans: an overview of their most important sourcing and applications as natural polysaccharides. Braz Arch Biol Technol 54:1075–1092

Distantina S, Wiratni, Fahrurrozi M, Rochmadi (2011) Carrageenan properties extracted from *Eucheuma cottonii*, Indonesia. World Acad Sci Eng Technol 54:738–742

Durako MJ, Dawes CJ (1980) A comparative seasonal study of two populations of *H. musciformis* from the east and west coasts of Florida, USA. I. Growth and chemistry. Mar Biol 59:151–156

FAO (2014) The state of world fisheries and aquaculture 2014. Opportunities and challenges FAO. Rome. p 233

Fournet I, Ar Gall E, Deslandes E, Huvenne J-P, Sombret B, Floc'h J-Y (1997) *In situ* measurements of cell wall components in the red alga *Solieria chordalis* (Solieriaceae, Rhodophyta) by FTIR microspectrometry. Bot Mar 40:45–48

Glenn E, Doty M (1990) Growth of the seaweeds *Kappaphycus alvarezii*, *K. striatum* and *Eucheuma denticulatum* as affected by environment in Hawaii. Aquaculture 84:245–255

Góes HG, Ries RP (2012) Temporal variation of the growth, carrageenan yield and quality of *Kappaphycus alvarezii* (Rhodophyta, Gigartinales) cultivated at Septiba Bay Southeastern Brazilian coast. J Appl Phycol 24:173–180

Hayashi L, Paula EJD, Chow F (2007) Growth rate and carrageenan analyses in four strain of *Kappaphycus alvarezii* (Rhodophyta, Gigartinales) farmed in subtropical waters of São Paulo State, Brazil. J Appl Phycol 19:393–399

Hu C, Feng L, Hardy RF, Hochberg EJ (2015) Spectral and spatial requirements of remote measurements of pelagic *Sargassum* macroalgae. Remote Sens Environ 167:229–246

Hung LD, Hori K, Nang HQ, Kha T, Hoa LT (2009) Seasonal changes in growth rate, carrageenan yield and lectin content in the red alga *Kappaphycus alvarezii* cultivated in Camranh Bay, Vietnam. J Appl Phycol 21:265–272

Huang B, Banzon VF, Freeman E, Lawrimore J, Liu W, Peterson TC, Smith TM, Thorne PW, Woodruff SD, Zhang HM (2015) Extended reconstructed sea surface temperature version 4 (ERSST.v4). Part I: upgrades and intercomparisons. J Climate 28:911–930

Huo Y, Han H, Shi H, Wu H, Zhang J, Yu K, Xu R, Liu D, Zhang Z, Liu K, He P, Ding D (2015) Changes to the biomass and species composition of *Ulva* sp. on *Porphyra* aquaculture rafts, along the coastal radial sandbank of the Southern Yellow Sea Yuanzi. Marine Poll Bull 93:210–216

Hurtado AQ, Gerung GS, Yasir SA, Critchley T (2013) Cultivation of tropical red seaweeds in the BIMP-EAGA region. J Appl Phycol 26: 707–718

Iskandar A, Syam R, Trijuno DD, Rahmi D (2014) Content of carrageenan, chlorophyll a and carotenoid of *Kappaphycus alvarezii* cultivated in different seawater depth Laikang village, District of Mangarabombang, Takalar Regency. J Appl Biotechnol 2:1–9

John DM, Asare SO (1975) A preliminary study of the variations in yield and properties of phycocolloids from Ghanaian seaweeds. Mar Biol 30:325–330

Kim SK (2014) Marine cosmeceuticals. J Cosmet Dermatol 13:56–67

Kloareg B, Quatrano RS (1988) Structure of cell walls of marine algae and ecophysiological functions of the matrix polysaccharides. Oceanogr Mar Biol Ann Rev 26:259–315

Laurienzo P (2010) Marine polysaccharides in pharmaceutical applications: an overview. Mar Drugs 8:2435–2465

Lazure P, Dumas F (2008) An external-internal mode coupling for 3D hydrodynamical model for applications at regional scale (MARS). Adv Water Res 31:233–250

Lechat H, Amat M, Mazoyer J, Gallant DJ, Buléon A, Lahaye M (1997) Cell wall composition of the carrageenophytes *Kappaphycus alvarezii* (Gigartinales, Rhodophyta) partitioned by wet sieving. J Appl Phycol 9:565–572

Mendoza WG, Ganzon-Fortes ET, Villanueva RD, Romero JB, Montano MNE (2006) Tissue age as a factor affecting carrageenan quantity and quality in farmed *Kappaphycus striatum* (Schmitz) Doty ex Silva. Bot Mar 49:57–64

MMAF (2015) Ministry of Marine Affairs and Fisheries http://statistik.kkp.go.id/

Mulyati H, Geldermann J (2017) Managing risks in the Indonesian seaweed supply chain. Clean Techn Environ Policy 19:175–189

Muñoz J, Freile-Pelegrín Y, Robledo D (2004) Mariculture of *Kappaphycus alvarezii* (Rhodophyta, Solieriaceae) color strains in tropical waters of Yucatán, México. Aquaculture 239:161–177

Nagao M, Matsuyama T (1979) Edge preserving smoothing. CGIP 9: 394–407

Necas J, Bartosikova L (2013) Carrageenan: a review. Vet Med 58:187–205

Oppelt N (2012) Hyperspectral classification approaches for intertidal macroalgae habitat mapping: a case study in Heligoland. Opt Eng 51:1–12

Orbita MLS (2013) Growth rate and carrageenan yield of *Kappaphycus alvarezii* (Rhodophyta, Gigartinales) cultivated in Kolambugan, Lanao del Norte, Mindanao, Philippines. Adv Agric Bot 5:128–139

Percival E (1979) The polysaccharides of green, red and brown seaweeds: their basic structure, biosynthesis and function. Brit Phycol J 4:103–117

Pereira L, Amado AM, Critchley AT, van de Velde F, Ribeiro-Claro PJA (2009) Identification of selected seaweed polysaccharides (phycocolloids) by vibrational spectroscopy (FTIR–ATR and FT-Raman). Food Hydrocoll 23:1903–1909

Periyasamy C, Subba Rao PV, Anantharaman P (2016) Spatial and temporal variation in carrageenan yield and gel strength of cultivated *Kappaphycus alvarezii* (Doty) Doty in relation to environmental parameters in Palk Bay waters, Tamil Nadu, Southeast coast of India. J Appl Phycol 28:525–532

Prajapati VD, Maheriya PM, Jani GK, Solanki HK (2014) Review: carrageenan: a natural seaweed polysaccharide and its applications. Carbohydr Polym 105:97–112

Radiarta IN, Saitoh SI, Yasui H (2011) Aquaculture site selection for Japanese kelp (*Laminaria japonica*) in southern Hokkaido, Japan, using satellite remote sensing and GIS-based models. ICES J Mar Sci 68:773–780

Radiarta N, Erlania I, Rusman D (2013) Impact of climate on seaweed, *Kappaphycus alvarezii* planting periods in Gerupuk Bay, Central Lombok Regency, West Nusa Tenggara. J Ris Akuakultur 8(3): 453–464

Reis RP, Yoneshigue-Valentin Y, Pereira dos Santos C (2008) Spatial and temporal variation of *Hypnea musciformis* carrageenan (Rhodophyta-Gigartinales) from natural beds in Rio de Janeiro State, Brazil. J Appl Phycol 26:1939–1951

Rhein-Knudsen N, Ale MT, Meyer AS (2015) Seaweed hydrocolloid production: an update on enzyme assisted extraction and modification technologies. Mar Drugs 13:3340–3359

Rochas C, Lahaye M, Yaphe W (1986) Sulfate content of carrageenan and agar determined by infrared spectroscopy. Bot Mar 29:335–340

Rossi N, Perrot T, Daniel C, Bajjouk T, Dion P (2010) Evolution temporelle de la couverture en Fucales sur les côtes bretonnes. Rebent 2010:13–14

Stanley NF (1987) Production properties and uses carrageenan. In McHugh (Ed) A guide to the seaweed industry. FAO Fish. Tech Paper No. 288, pp. 188–123

Stengel DB, Connan S, Popper ZA (2011) Algal chemodiversity and bioactivity: sources of natural variability and implications for commercial application. Biotechnol Adv 29:483–501

Stiger-Pouvreau V, Bourgougnon N, Deslandes E (2016) Carbohydrates from seaweeds. In: Fleurence J, Levine I (eds) Seaweed in health and disease prevention. Elsevier, Amsterdam, pp 223–274

Stiger V, Payri CE (2005) Natural settlement of a young population of *Turbinaria ornata* and phenological comparisons with older populations. Aquat Bot 81:225–243

Susanto RD, Marra J (2005) Effects of the 1997/98 El Niño on chlorophyll a variability along the southern coasts of Java and Sumatra. Oceanography 18:124–127

Susanto RD, Moore TS II, Marra J (2006) Ocean color variability in the Indonesian seas during the SeaWiFS era. Geochem Geophys Geosyst 7:1–16

Tanniou A, Vandanjon L, Gonçalves O, Kervarec N, Stiger-Pouvreau V (2015) Rapid geographical differentiation of the European spread brown algae *Sargassum muticum* using HRMAS NMR and Fourier-transform infrared spectroscopy. Talanta 132:451–456

Utojo, Mansyur A, Pantjara B, Marsambuana, Pirzan, Hasnawi (2007) The suitable waters environment condition of Malasoro Bay for *Eucheuma* sp. seaweed culture development location. J Ris Akua 2:243–255

Vairappan CS (2006) Seasonal occurrences of epiphytic algae on the commercially cultivated red alga *Kappaphycus alvarezii* (Solieriaceae, Gigartinales, Rhodophyta). J Appl Phycol 18:611–617

Wenno PA, Syamsuddin R, Zainuddin EN, Ambo-Rappe R (2015) Cultivation of red seaweed *Kappaphycus alvarezii* (Doty) at different depths in South Sulawesi, Indonesia. Bioflux 8:468–473

Wijesekara I, Kim SK (2015) Application of marine algae derived nutraceuticals in the food industry in marine algae extracts. In: Chojnacka K, Kim SK (eds) Marine algae extracts: processes, products, and applications 2. Wiley-VCH, Weinheim, pp 627–638

Zablackis E, Vreeland V, Dodoszewski B, Laetsch WM (1991) Differential localization of carrageenan gelling sequences in *Kappaphycus alvarezii* var. *tambalang* (Rhodophyta) with FITC-conjugated carrageenan oligosaccharides. J Phycol 27:241–248

Zamroni A, Laoubi K, Yamao M (2011) The development of seaweed farming as a sustainable coastal management method in Indonesia: an opportunities and constraints assessment. In: Anderson R, Beriatos E (eds) Development and planning. WIT Press, Southampton, UK, pp 505–516

Zubia M, Andréfouët S, Payri C (2015) Distribution and biomass evolution of drifting brown algae from Moorea lagoon (French Polynesia) for eco-friendly agricultural use. J Appl Phycol 27:1277–1287